Calculations for the preferred design (Design A): carousel pill dispenser

1. Assumptions

- Sizes and masses come from the CAD model (models/esp2025.py): all 37 parts are modelled with their own material density; the dispenser has 54 parts in all.
- The largest pill in the brief: a cylinder 5 mm across and 20 mm long, 1.3 g/cm³ (0.51 g).
- Twice a day dosing is the worst case for the battery. The patient presses the button within 3 minutes of the alarm on average.
- Labour £17 an hour (brief). Material and bought in prices are supplier estimates for batches of 1500 to 3000; moulding at £28 an hour of machine time with its operator share.
- Friction: pills and PP carousel on the PP deck μ = 0.3; acetal worm on acetal gear μ = 0.25.
- Prying force on the lid 200 N: a strong two handed pull at the front edge by someone trying to get at the pills.

2. Size and mass against the brief

Check	Working	Result
Length (front of the drawer to the back of the hinge)	CAD bounding box (brief: 190 max)	**189.0 mm**
Width	case Ø180 (brief: 190 max)	**180.0 mm**
Height (feet to the top of the button)	CAD bounding box (brief: 60 max)	**58.6 mm**
Mass with the batteries, empty	sum of volume × density for all 54 parts; cells 45 g	**413 g**
Mass when full of the largest pills	413 + 43 g of pills (3 in each of 28 compartments)	456 g

Sub-assembly	Mass
Chassis, pharmacy lock and chute	81 g
Carousel and drive gear	68 g
Base: bottom cover, batteries and drive	134 g
Control pod: display, PRESS HERE button, light and alarm	56 g
Lid and hinge	68 g
Dispensing drawer	6 g

The brief limits the unit with its batteries: 413 g leaves 87 g in hand. The walls are 1.8 to 2 mm of polypropylene, the lightest common moulding plastic (905 kg/m³), and four AAA cells (45 g) were chosen in place of AA cells to save about 50 g.

3. Doses and pill capacity

Quantity	Working	Result
Compartments	30 segments of 12°, less 2 capped blanks	28
Once a day medication	28 compartments ÷ 1 a day (brief: refill every 28 days)	**28 days**
Twice a day medication	28 ÷ 2 a day (brief: refill every 14 days)	**14 days**
Row 1: pills from radius 41.5 to 61.5	gap between ribs at the inner end 7.7 mm / 5 mm	1 a layer
Layers	depth 18.5 mm (a 1.5 mm gap kept under the lid) ÷ 5 mm	3
Largest pills in one compartment	1 a layer × 3 layers (brief: at least 3)	**3**
Volume of one compartment	$(\pi(R_0^2 - R_i^2) / 30 - \text{rib}) \times \text{depth}$	8.4 cm³
Gap under the lid	52 − 50 (smaller than a pill, so pills cannot climb into the next compartment)	2.0 mm

4. Battery life (4 × AAA, 6 V)

Load	Current	Time a day	Charge a day	How it is used
Microcontroller asleep with the clock running	0.004 mA	86400 s	0.096 mAh	all day
Display (reflective LCD, always on)	0.012 mA	86400 s	0.288 mAh	all day, no backlight
Gear motor turning the carousel	120 mA	3.2 s	0.107 mAh	2 doses, 4 turns of the worm at 150 rpm = 1.6 s each
Alarm: buzzer and flashing light, until the button is pressed	30 mA	360 s	3.000 mAh	2 doses; allow 3 minutes each before the patient presses
Microcontroller awake (dose, display update, sensors)	3 mA	68 s	0.057 mAh	10 s a dose and 2 s an hour
Home and drop sensors (infrared)	20 mA	6 s	0.033 mAh	only while the carousel turns and the pills fall
NFC programming by the pharmacy	0 mA	0 s	0.000 mAh	powered by the phone or tablet, not the battery
Total a day, twice a day dosing			**3.58 mAh**	

Quantity	Working	Result
Usable charge	1100 mAh × 0.8 (cells in series: the same current flows through all four)	880 mAh
Battery life	880 / 3.58	**246 days**
Refills on one set of cells (twice a day)	246 / 14	18

The pharmacy changes the cells at a refill when the display shows the low battery sign (at 20 % left there are still about 61 days). The clock keeps time for 2 minutes without cells, long enough to change them.

5. The drive: worm and wheel

Quantity	Working	Result
Carousel and gear (CAD)	sum of volume × density	68 g
Pills, every compartment full	3 × 28 × 0.51 g	43 g
Torque to turn the carousel	μ m g r = 0.3 × 0.111 × 9.81 × 0.0597	19.4 N mm
Gear ratio	120 teeth, single start worm	120:1
Torque at the motor	19.4 / 120 / 0.4 (efficiency)	0.41 N mm
Safety factor on the motor	stall 98 N mm / 0.41: it never struggles, even with a stuck pill	**242**
Worm turns a dose	12° / (360 / 120)° a tooth	4
Time to turn one dose	4 turns at 150 rpm	1.6 s
Can the carousel be forced round by hand?	worm lead angle $\tan^{-1}(2.513 / \pi \times 5.8) = 7.9°$ < friction angle $\tan^{-1}0.25 = 14.0°$	**No: self locking**

6. Tamper-proof lock: the lid catch

Quantity	Working	Result
Prying force on the lid, all taken by the catch hook	assumption (the hinge takes the rest of the lid)	200 N
Direct stress in the tab	200 / (19.5 × 3)	3.4 MPa
Bending in the tab: the cam presses on the hook 2.50 mm out from the middle of the tab	200 × 2.50 / (19.5 × 3² / 6)	17.1 MPa
Total in the tab (outside face)	3.4 + 17.1	20.5 MPa
Hook bending and shear at its root	200 × 1.0 / (19.5 × 3.5² / 6); 200 / (19.5 × 3.5)	5.0; 2.9 MPa
Safety factor (PP-GF30, 75 MPa)	75 / 20.5	**3.7**
FEA (PP-GF30, 200 N): peak von Mises 23.3 MPa, safety factor 3.2. The FEA also shows the stress round the inside corner of the hook, where the hand calculation has no stress concentration. The lock cam is steel and the lock body zinc: both are much stronger than the plastic catch, so the catch sets the limit.		

7. Waterproof and tamper-proof: how every way in is closed

Opening	How it is sealed	Who can open it
Lid to chassis	Silicone gasket ring (item 33), clamped by the catch	Pharmacy key only
Lid round the control pod	O-ring (item 22) in the pod column, inside the lid collar	–
Button, light, display window	Each has a flange clamped under the pod cap; the cap is welded to the column	Nobody (sealed for life)
Bottom cover to chassis	The same gasket ring, squeezed by four screws	Factory
Battery doors	Silicone seals (item 18), pin Torx security screws	Pharmacy (bit kept at the counter)
Drawer	Silicone frame seal on the drawer front, pressed on the case when it clicks home	Patient (empty except the dose)
Programming	NFC through the plastic: no socket, no hole	Pharmacy app

Target IP67 (30 minutes under 1 m of water); each unit gets an air leak test at the end of the line. Only the compartment over the drop hole is ever open to the drawer, the worm cannot be back driven, and the lid needs the pharmacy key, so the patient can only ever take the dose that is due.

8. Cost of one unit, and the most units each budget can buy

Cost a unit	Working	£
Materials for the moulded parts	mass × price × buy to fly, every part (table 11)	£0.68
Moulding	cycle seconds × £28 an hour	£4.68
Bought in parts (electronics, lock, cells, screws)	supplier prices for 1500+	£15.82
Assembly and test labour	(14 + 6) min × £17 an hour	£5.67
Packaging and instructions		£0.90
Overhead and margin	15 % of £27.75	£4.16
Cost a unit before the one-off costs		**£31.91**
One-off: mould tools and design, prototypes and tests	£31,000 + £9,000, shared by the order	£40,000

Order	Budget	Cost a unit	Against	Order cost	Most units the budget can buy
1500 units	£97,500	£31.91 + £40,000 / 1500 = **£58.58**	**£65 target**	£87,868	(£97,500 − £40,000) / £31.91 = **1,801**
3000 units	£120,000	£31.91 + £40,000 / 3000 = **£45.25**	**£40 target**	£135,735	(£120,000 − £40,000) / £31.91 = **2,506**

The 1,500 order comes in under its £65 target. The 3,000 order is over its £40 target, so that budget buys fewer units (right hand column). The £40 target for 3000 units is the hard one: the electronics (£11.58) are the biggest cost, so the design uses one motor and one drop hole in place of a motor or latch for each compartment.

9. Programme

Task	Start week	End week	Weeks
Design and drawings (this task), health trust approval	0	4	4
Prototypes 3D printed, user trial with patients and pharmacists	3	9	6
Mould tools made and first parts sampled	6	16	10
Tests: IP67 water, 1 m drop, 10,000 dose cycles, EMC; changes	14	18	4
Production line set up, pilot batch of 50	17	19	2
Production of 1500 units: 1 line at 150 a week, then delivery	19	30	11
Production of 3000 units: 2 lines at 300 a week, then delivery	19	30	11

1500 units are delivered in week 30 (brief: 30 weeks); 3000 units, with a second assembly line, in week 30 (brief: 35 weeks).

10. Materials and recycling

Material	£/kg	Buy to fly	Why
Polypropylene homopolymer, white or blue	£1.60	1.08	Recyclable (code 5)
Clarified polypropylene, clear	£2.10	1.08	Recyclable (code 5); clear like glass
PP with 30 % glass fibre	£3.20	1.10	Stiff and strong for the catch
Acetal (POM)	£3.80	1.10	Low friction gears, no grease
Silicone rubber	£12.00	1.15	Seals that stay soft for years
TPE feet	£6.00	1.10	Bought as a reel of self adhesive feet

All the case parts are polypropylene: 225 g of the 413 g. With the clear lid and carousel, 69 % of the mass is recyclable polypropylene (code 5) and needs no separating; the boards, cells and lock come out with eight screws for WEEE recycling.

11. Mass and cost of every part (one dispenser)

Item	Part	Qty	Mass g	Material	Material	Moulding	Bought in
1	chassis	1	69.5	Polypropylene homopolymer, white or blue	£0.12	£0.31	–
2	pharmacy cam lock	1	8.0	Bought in	–	£0.16	£1.90
3	lock cam	1	1.9	Bought in	–	£0.04	£0.15
4	dispensing chute	1	1.0	Polypropylene homopolymer, white or blue	–	£0.12	–
5	drop sensor (infrared)	2	0.3	Bought in	–	£0.23	£0.60
6	carousel	1	55.4	Polypropylene homopolymer, white or blue	£0.10	£0.27	–
7	drive gear (worm wheel)	1	12.3	Acetal (POM)	£0.05	£0.14	–
8	bottom cover	1	48.6	Polypropylene homopolymer, white or blue	£0.08	£0.23	–
9	rubber foot	4	1.4	TPE feet	£0.01	£0.09	–
10	main circuit board	1	4.2	Bought in	–	£0.23	£5.20
11	battery holder 2 x AAA	2	9.5	Bought in	–	£0.16	£0.70
12	AAA cell	4	44.7	Bought in	–	£0.09	£1.00
13	gear motor 6 V	1	11.5	Bought in	–	£0.12	£2.20
14	motor clamp	1	0.5	Polypropylene homopolymer, white or blue	–	£0.06	–
15	worm	1	0.3	Acetal (POM)	–	£0.08	–
16	home sensor	1	0.1	Bought in	–	£0.08	£0.40
17	battery door	2	7.7	Polypropylene homopolymer, white or blue	£0.01	£0.16	–
18	battery door seal	2	1.7	Silicone rubber	£0.02	£0.08	–
19	security screw M2.2 x 8	4	1.4	Bought in	–	£0.25	£0.24
20	cover screw 2.5 x 10	4	2.2	Bought in	–	£0.25	£0.12
21	pod column	1	27.7	Polypropylene homopolymer, white or blue	£0.05	£0.14	–
22	pod O-ring	1	1.8	Bought in	–	£0.04	£0.08
23	panel circuit board	1	9.7	Bought in	–	£0.12	£1.10
24	display	1	4.9	Bought in	–	£0.08	£1.60
25	button switch	1	0.2	Bought in	–	–	£0.08
26	flashing light (LED)	1	0.1	Bought in	–	–	£0.05
27	light lens	1	0.2	Clarified polypropylene, clear	–	£0.06	–
28	buzzer	1	0.8	Bought in	–	–	£0.35
29	PRESS HERE button	1	2.6	Silicone rubber	£0.04	£0.08	–
30	display window	1	1.0	Clarified polypropylene, clear	–	£0.06	–
31	pod cap	1	7.3	Polypropylene homopolymer, white or blue	£0.01	£0.16	–
32	lid	1	58.2	Clarified polypropylene, clear	£0.13	£0.27	–
33	gasket ring	2	1.9	Silicone rubber	£0.03	£0.12	–
34	lid catch	1	3.1	PP with 30 % glass fibre	£0.01	£0.09	–
35	hinge pin	1	5.0	Bought in	–	£0.12	£0.05
36	dispensing drawer	1	5.9	Polypropylene homopolymer, white or blue	£0.01	£0.14	–
37	drawer seal	1	0.1	Silicone rubber	–	£0.06	–